NYC Airbnb Price Prediction - Linear Regression

Minor Project - One-Page Summary

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Dataset	NYC Airbnb Open Data (48,895 listings)	**Target**	Nightly price ($)

Objective

Predict the nightly price of a New York City Airbnb listing with standard multiple Linear Regression while completing the required EDA, cleaning, outlier treatment, feature encoding, train-test validation, evaluation and residual analysis.

Data Preparation

Missing values: *last_review* and *reviews_per_month* each have 10,052 missing values (20.56%). *last_review* was dropped. Missing *reviews_per_month* values occur for listings with zero reviews, so they were filled with 0. Identifiers and free-text labels (*id, name, host_id, host_name*) were removed.

Outliers and features: Values above the 95th percentile were trimmed: price above $355 and minimum stays above 30 nights. The 200+ level *neighbourhood* field was dropped to avoid excessive dimensionality. *neighbourhood_group* and *room_type* were one-hot encoded with one reference level dropped. The final data was split 80%/20% using random_state=42.

Model Results

Metric	Result	Meaning
R-Squared (R^2)	0.4708	About 47.1% of nightly-price variation is explained.
MAE	$38.54	Average prediction misses the true nightly price by about $38.54.
MSE	2834.96	Large mistakes receive heavier penalties because errors are squared.
RMSE	$53.24	Error scale after giving larger misses extra weight.

Business Interpretation

For the typical-market listings retained after trimming, the model is a useful baseline rather than an exact pricing engine. An MAE of $38.54 means a suggested price should be viewed as an approximate nightly estimate.

Cheap vs expensive: the model performs better on cheaper and mid-priced listings. Premium listings have larger errors because luxury amenities, views, interior quality, special events and other nonlinear factors are not captured by the available columns.

Conclusion & Limitation

The project satisfies the requested Linear Regression workflow. The model uses 2019 data, intentionally removes extreme listings and omits many qualitative or seasonal price drivers, so it is appropriate for learning and benchmarking rather than live production pricing.